

KUANDIAN, China

WMO: 544930

Lat: 40.7088N

Lon: 124.7817E

Elev: 856

StdP: 14.25

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-3.5	-17.1	2.2	0.6	-13.0	2.8	3.1	13.7	15.3	12.1	16.4	1.5	200	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	86.9	72.7	84.2	71.6	81.9	70.7	76.8	82.2	75.4	79.8	74.2	78.2	5.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	137.1	79.6	74.1	131.5	78.1	72.9	126.2	76.7	41.0	82.4	39.7	79.8	38.5	78.2	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	11.5	9.8	-12.2	91.1	5.4	2.6	-16.1	93.0	-19.2	94.5	-22.2	95.9	-26.1	97.8
			-12.6	78.5	5.3	2.1	-16.4	80.1	-19.5	81.3	-22.5	82.5	-26.3	84.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.2	13.6	21.0	33.5	47.5	59.0	68.0	73.9	73.1	63.2	49.4	33.1	18.1	(6)
(7)		DBStd	22.01	8.17	8.59	7.25	7.04	5.29	4.23	3.52	4.24	5.49	7.07	9.41	8.56	(7)
(8)		HDD50	4180	1128	812	511	128	4	0	0	1	96	510	990	(8)	
(9)		HDD65	7537	1593	1232	976	525	196	19	0	3	97	484	957	1455	(9)
(10)		CDD50	2808	0	0	1	53	281	539	741	718	395	77	3	0	(10)
(11)		CDD65	690	0	0	0	0	9	108	276	256	41	0	0	0	(11)
(12)		CDH74	4399	0	0	0	6	170	772	1567	1615	263	6	0	0	(12)
(13)		CDH80	962	0	0	0	0	17	170	393	359	23	0	0	0	(13)
(14)	Wind	WSAvg	3.5	2.9	3.6	4.3	4.6	4.2	3.4	3.1	3.1	3.0	3.2	3.3	3.0	(14)
(15)	Precipitation	PrecAvg	43.3	0.4	0.6	0.7	2.1	2.9	4.2	12.4	11.9	3.7	2.0	1.5	0.6	(15)
(16)		PrecMax	71.5	2.0	1.5	1.9	4.9	6.6	8.3	34.8	24.6	8.7	4.6	3.3	1.5	(16)
(17)		PrecMin	26.0	0.0	0.0	0.1	0.4	0.7	1.2	1.9	0.9	1.3	0.0	0.2	0.0	(17)
(18)		PrecStd	10.2	0.4	0.4	0.5	1.3	1.5	1.9	6.7	6.2	1.7	1.1	0.8	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	47.9	59.6	74.6	82.7	89.0	91.3	89.7	82.8	73.5	59.0	41.4	(19)
(20)		MCWB	32.0	37.8	44.5	55.0	61.6	68.8	76.3	75.3	69.3	60.2	50.9	36.9	(20)	
(21)		2%	DB	33.9	41.7	53.7	69.9	78.6	84.6	87.3	86.1	79.4	68.8	54.9	37.6	(21)
(22)		MCWB	29.5	34.2	41.8	53.0	59.7	66.8	74.2	73.7	66.9	56.7	49.0	34.0	(22)	
(23)		5%	DB	30.8	38.1	49.6	66.0	75.5	81.4	84.2	83.7	76.9	65.3	51.3	34.7	(23)
(24)		MCWB	26.9	32.1	39.0	50.2	58.8	66.3	73.1	72.6	64.5	54.4	46.0	31.4	(24)	
(25)		10%	DB	27.7	34.9	46.0	62.1	72.4	78.5	81.6	81.6	74.6	61.9	47.9	31.9	(25)
(26)		MCWB	23.9	30.2	37.3	48.5	57.1	65.7	72.5	72.0	63.3	52.9	43.1	28.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.0	39.5	47.3	59.7	66.5	72.2	79.7	78.6	72.7	62.5	53.7	38.8	(27)
(28)		MCDWB	36.3	45.4	54.3	66.9	74.8	80.2	87.4	84.5	78.2	69.5	57.0	40.5	(28)	
(29)		2%	WB	30.4	35.5	43.5	55.4	63.3	70.5	77.3	77.0	70.3	59.6	50.2	34.7	(29)
(30)		MCDWB	33.3	40.4	50.8	65.9	73.0	78.2	83.3	82.2	75.7	65.6	53.9	37.0	(30)	
(31)		5%	WB	27.3	33.0	40.8	52.6	61.2	69.1	75.5	75.8	68.1	57.3	47.1	31.8	(31)
(32)		MCDWB	30.5	37.0	47.1	62.4	70.6	76.6	80.2	80.2	73.8	62.3	50.3	34.1	(32)	
(33)		10%	WB	24.1	30.7	38.4	50.2	59.5	67.8	74.3	74.7	66.2	55.0	43.7	28.9	(33)
(34)		MCDWB	27.4	34.9	45.0	59.2	68.5	75.2	78.4	78.6	71.1	59.8	47.0	31.6	(34)	
(35)	Mean Daily Temperature Range		MDBR	21.3	20.3	18.7	21.7	21.5	17.2	12.7	14.9	20.7	21.6	17.4	18.7	(35)
(36)		5% DB	MCDBR	21.4	21.7	24.7	28.9	28.4	24.8	18.8	18.6	22.7	25.7	20.4	19.0	(36)
(37)			MCWBR	17.0	14.6	13.9	13.9	12.1	9.6	7.3	7.7	10.6	13.8	13.7	15.0	(37)
(38)			MCDBR	19.3	18.3	21.5	23.4	22.4	17.6	12.8	12.5	17.0	20.1	16.5	16.7	(38)
(39)		5% WB	MCWBR	16.1	13.2	13.8	12.7	10.8	8.0	5.9	6.2	9.0	12.6	12.7	14.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.374	0.450	0.513	0.598	0.630	0.701	0.668	0.574	0.504	0.529	0.456	0.386	(40)	
(41)		taud	1.984	1.828	1.757	1.649	1.625	1.558	1.673	1.869	1.970	1.807	1.902	1.988	(41)	
(42)		Ebn at Noon	232	229	230	221	218	203	208	225	232	202	199	214	(42)	
(43)		Edh at Noon	39	54	65	77	81	86	76	61	52	55	43	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	773	1016	1354	1541	1741	1554	1321	1401	1377	1030	711	640	(44)	
(45)		RadStd	42	67	89	111	135	179	141	162	115	64	81	50	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-1.69	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page